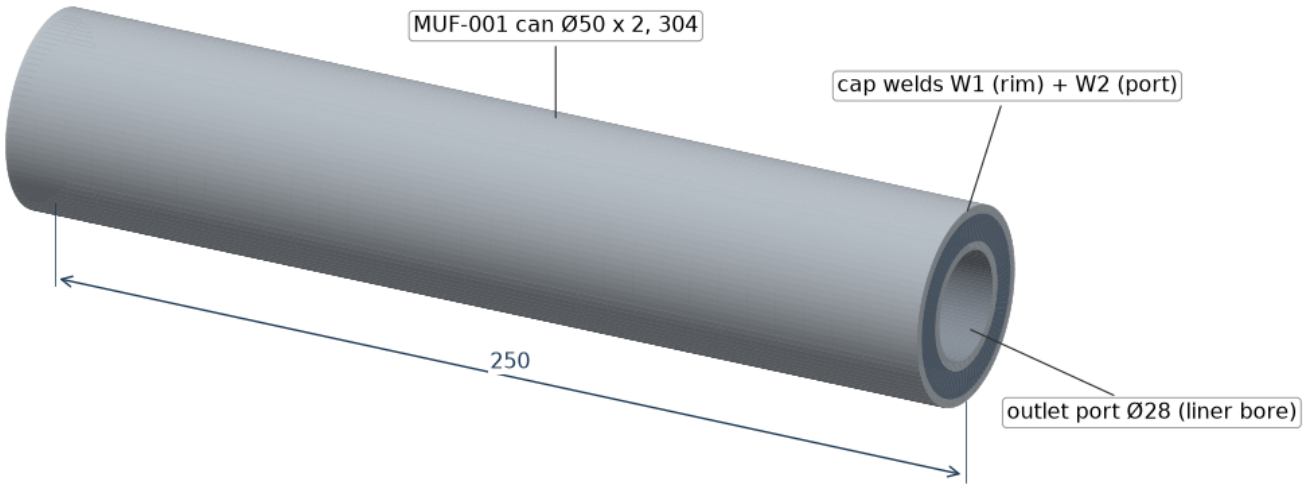
50 x 250 mm Baffle Muffler

Build Book - home TIG shop



Item	Spec
Outer can	Ø50 x 2 mm wall x 250 mm, 304
Liner	Ø32 x 2 mm wall x 250 mm, 304 tube (bought, no seam)
Perforation	112 x Ø9 mm, 8/row, 14 rows @ 12 mm, 22.5° stagger; 28.4% open (OD basis) / 32.4% (ID basis)
Caps / baffles	3 mm 304 discs: 2 caps + 4 baffles @ 50, 100, 150, 200 mm; OD = measured can ID - 0.2
Welding	TIG DCEN / pulse, ER308L 1/16 in, max 100 A (machine 110 A, 220 V)
Material	AISI 304 throughout (316/316L + ER316L for salt/marine)

Contents: 1 How to Build - 2 Drawings - 3 3D Views - 4 TIG Parameters - 5 BOM + Sourcing - 6 Verification Checklist + Design Checks

1. How to Build

Work top to bottom. Each step lists the exact tools. Settings for every weld are in section 4.

Step 1. Measure

- Can ID at 4 clock positions, both ends -> smallest value. Liner OD likewise -> largest.
- Re-run `muffler.py --can-id <ID> --liner-od <OD>`; discs become OD = ID - 0.2, bore = OD + 0.2.

Tools: Calipers with ID jaws (0.01 mm), laptop with muffler.py

Step 2. Cut

- Can and liner to 250 +/-0.5, ends square within 0.3 (check with a square). Deburr.
- Order/cut discs: 2x MUF-003 caps, 4x MUF-004 baffles. Cut the JIG-001 guide plate.

Tools: Bandsaw or abrasive cut-off saw, machinist square, flat file, deburr tool, marker

Step 3. Jig - build the centering sleeve per `jig_build.md`.

Tools: Drill press, V-block, Ø4 mm cobalt drill, TIG/MIG for tacking block + M6 nut, M6 thumb screw, file

Step 4. Drill (liner)

- Wrap guide plate, edge flush with the inlet end, tape tight. Center punch all 112 pilot holes. Remove.
- Sleeve on liner, Ø4 pin through guide bushing into each punch mark, lock thumb screw, drill Ø4.
- Open each to Ø9 with step drill, liner in V-block. ~400 rpm for Ø9 in 304, cutting fluid, steady feed (304 work-hardens if the drill rubs).
- Deburr OD (file/flap wheel) and ID (round file, brush). Blow out chips.

Tools: Drill press, V-block, centre punch + hammer, Ø4 mm pin, Ø4 mm cobalt drill, step drill to Ø9 (or Ø9 cobalt), cutting fluid, round file, deburr tool

Step 5. Clean - acetone on all parts; stainless-only brushes; no grinding discs used on carbon steel.

Tools: Acetone, lint-free rags, stainless-only wire brush, nitrile gloves

Step 6. Fit

- Slide baffles onto liner at z = 50, 100, 150, 200 (centres) using spacer blocks (48.5 mm from the end, 47 mm between baffles). Baffles sit on undrilled bands.
- Dry-fit liner + baffles into can, caps in both ends: caps flush, nothing binds.

Tools: Spacer blocks 48.5 mm and 47 mm, square, calipers

Step 7. Weld W3 (baffles to liner)

- Foil-tape both liner ends, argon purge the bore 4-6 L/min.
- Check square, 4 tacks per baffle @ 90 deg, re-check square, continuous fillet on the upstream face.

Tools: TIG (DCEN/pulse), argon + purge line, foil tape, ER308L 1/16 in, stainless brush

Step 8. Core fill (optional, for a quieter can)

- Wrap stainless wool over each perforated band, then basalt/E-glass packing up to baffle OD (~30 g total).
- Hold with a couple of turns of masking tape (it burns off).

Tools: Stainless wool, basalt or E-glass packing, masking tape, gloves + dust mask (fibres)

Step 9. Fit can - slide the liner/baffle core into the can; the packing compresses. Ends flush.

Tools: Rubber mallet, gloves

Step 10. Cap

- Press caps in flush both ends. Purge the liner bore from the far end (perforations purge the chambers).
- 4 tacks each joint, then W2 (cap to liner) and W1 (cap to can rim), alternating ends to limit distortion.

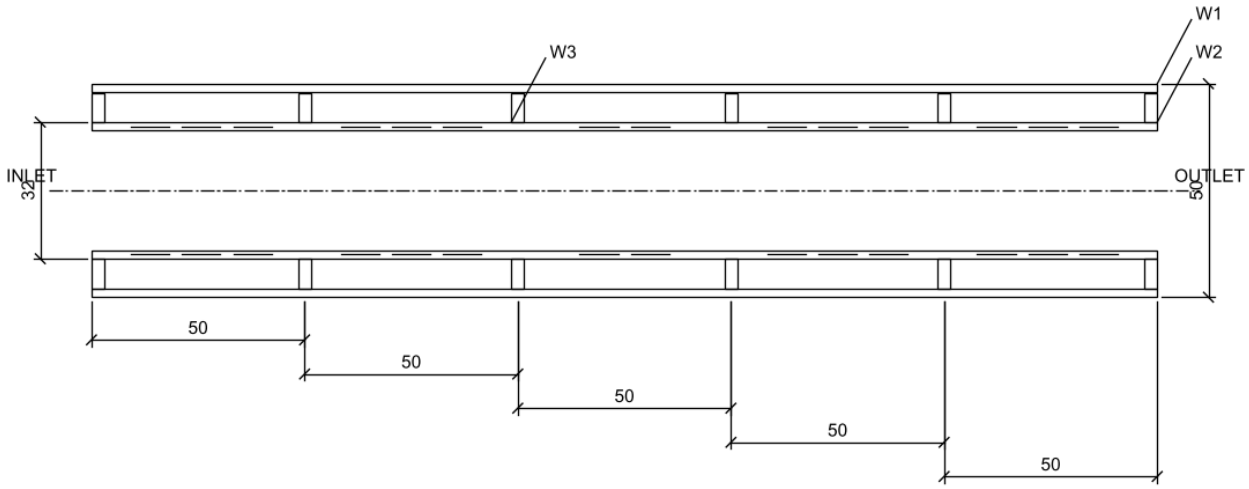
Tools: TIG, argon purge, foil tape, clamps, ER308L 1/16 in

Step 11. Finish - pickle/passivate welds; run `verification_checklist.md`.

Tools: Pickling paste (acid: gloves + eye protection), Schrader-valve plug, expansion plug, soapy water, brush

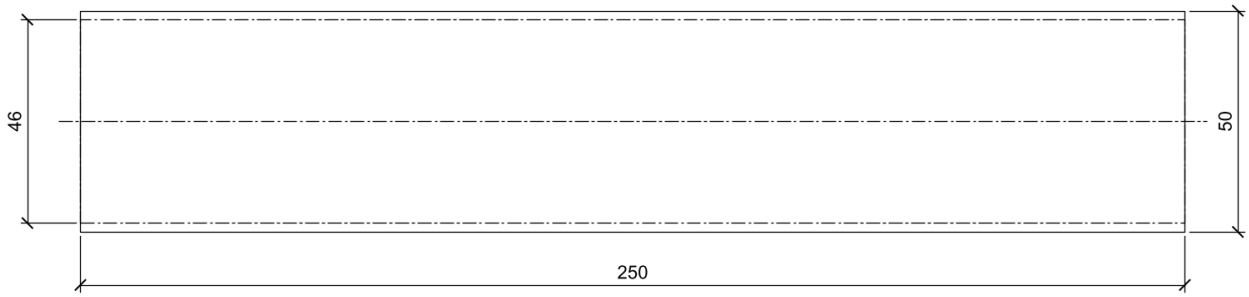
2. Drawings

Full-size DXF files are in the package for laser/print; these are previews. Units: mm.



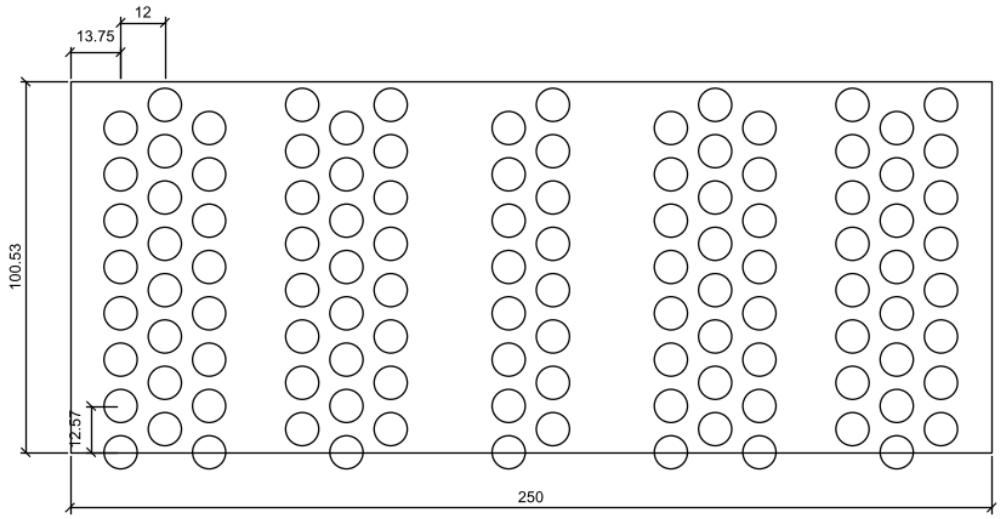
MUF-000 ASSEMBLY HALF-SECTION
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER = do not cut
W1 cap-can rim | W2 cap-liner port | W3 baffle-liner (see weld map)
Optional packing (core fill) in annulus between baffles

MUF-000 Assembly half-section: can, liner, caps, 4 baffles; weld callouts W1-W3.



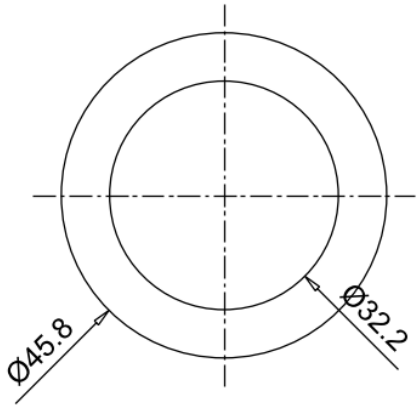
MUF-001 OUTER CAN
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER = do not cut
TUBE 50 OD x 2 WALL x 250 LG, ENDS SQUARE WITHIN 0.3, DEBURR
MEASURE ID AT 4 CLOCK POSITIONS BOTH ENDS; USE SMALLEST FOR DISC OD

MUF-001 Outer can: 50 OD x 2 wall x 250 tube, ends square, measure ID x4.



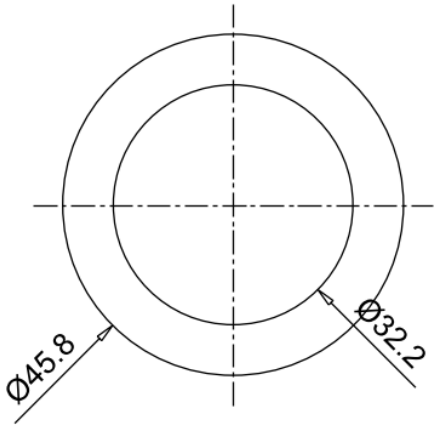
MUF-002 PERFORATED LINER - WRAP LAYOUT (OD)
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER = do not cut
PURCHASED TUBE 32 OD x 2 WALL x 250 LG, 304 (seamless preferred)
112 HOLES DIA 9 THRU RADIAL, 14 ROWS, 8/ROW @ 45 deg
ALTERNATE ROWS ROTATED 22.5 deg (HALF PITCH STAGGER). Y = arc on OD from datum line
OPEN AREA 28.4% (OD BASIS). DEBURR ID AND OD. DRILL WITH JIG (drill_jig.dxf)

MUF-002 Liner wrap layout (OD): 112 x Ø9 holes, 8/row, 14 rows, 22.5° stagger.



MUF-003 END CAP
MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER = do not cut
THK 3 sheet, QTY 2
OD = MEASURED CAN ID - 0.2; BORE = MEASURED LINER OD + 0.2
Bore = inlet/outlet port
Fits flush inside can ends

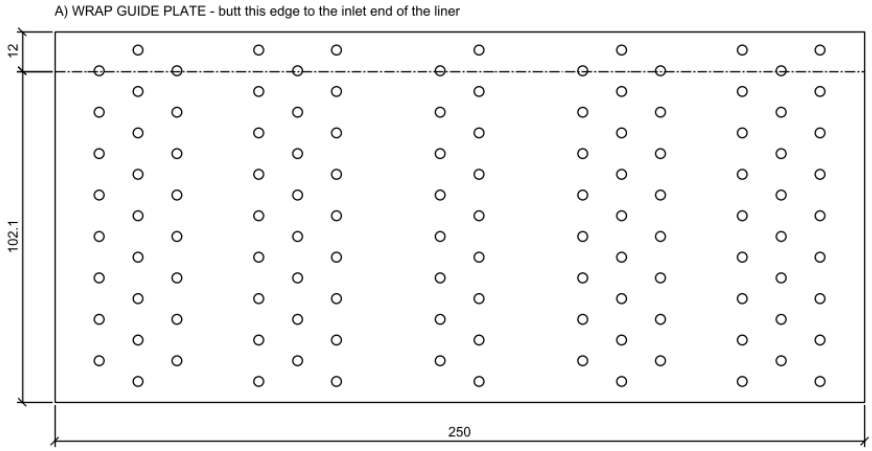
MUF-003 End cap (x2): 3 mm disc, OD = can ID - 0.2, bore = liner OD + 0.2 (port).



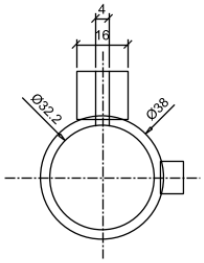
MUF-004 BAFFLE

MATL: AISI 304 (316 for corrosive/salt service)
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER = do not cut
THK 3 sheet, QTY 4
OD = MEASURED CAN ID - 0.2; BORE = MEASURED LINER OD + 0.2
Stations z = 50, 100, 150, 200

MUF-004 Baffle (x4): same 3 mm blank as the cap, stations at 50/100/150/200.



0.5 mm shim (Al/steel) or 1:1 print glued to card. Pilot holes DIA 3.
Tab overlaps start; tab holes coincide when wrapped. Circumference at shim mid = 102.10



B) CENTERING SLEEVE (end view)

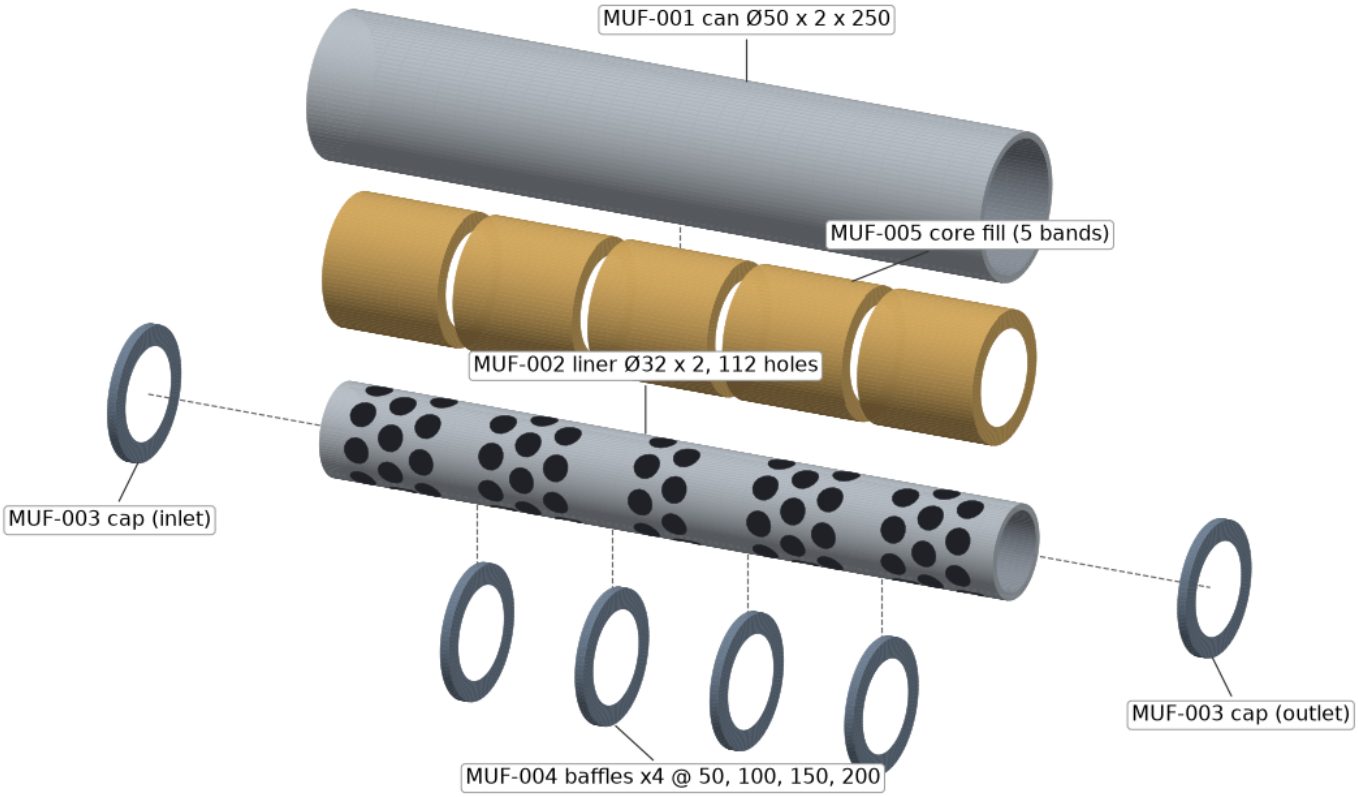
Sleeve: 38 OD tube, bore 32.2, 40 long
Bushing block 16x15x40 steel, guide hole DIA 4 RADIAL through axis; M6 nut + thumb screw on side to lock

JIG-001 DRILL JIG

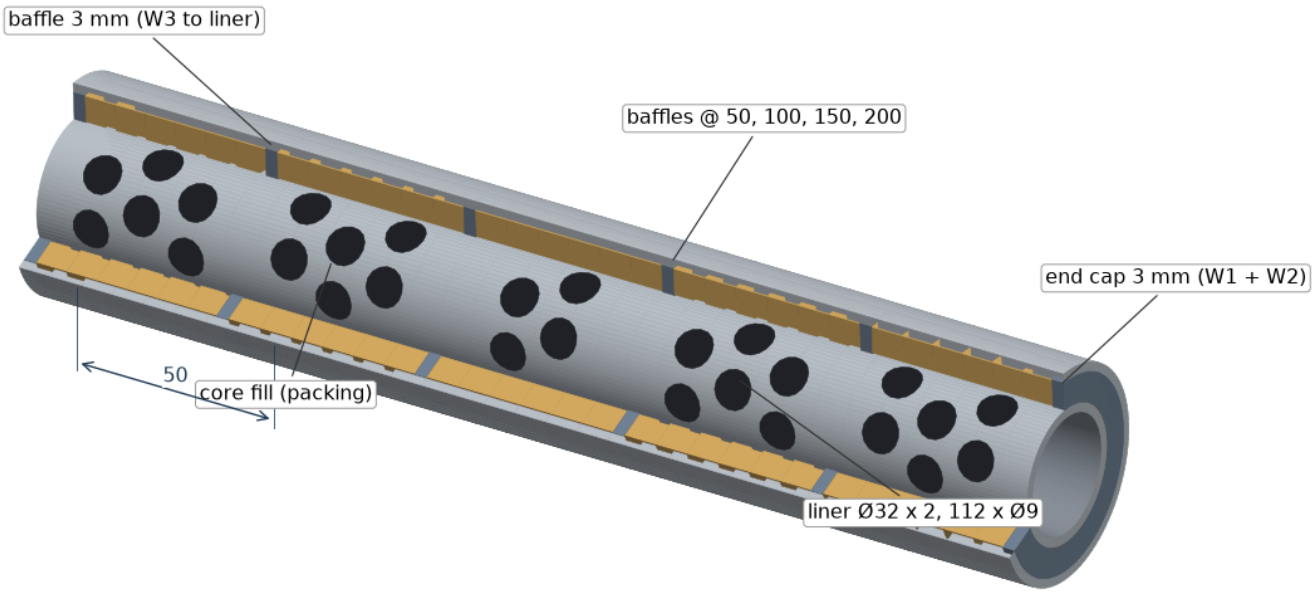
MATL: shim: 0.5 Al/steel; sleeve + block: mild steel
UNITS: mm TOL: +/-0.2 unless noted
LAYERS: CUT/HOLES = cut; DIM/TEXT/CENTER = do not cut
112 pattern positions = liner pattern (112)

JIG-001 Drill jig: 0.5 mm wrap-around shim (112 pilots) + centring sleeve with Ø4 guide.

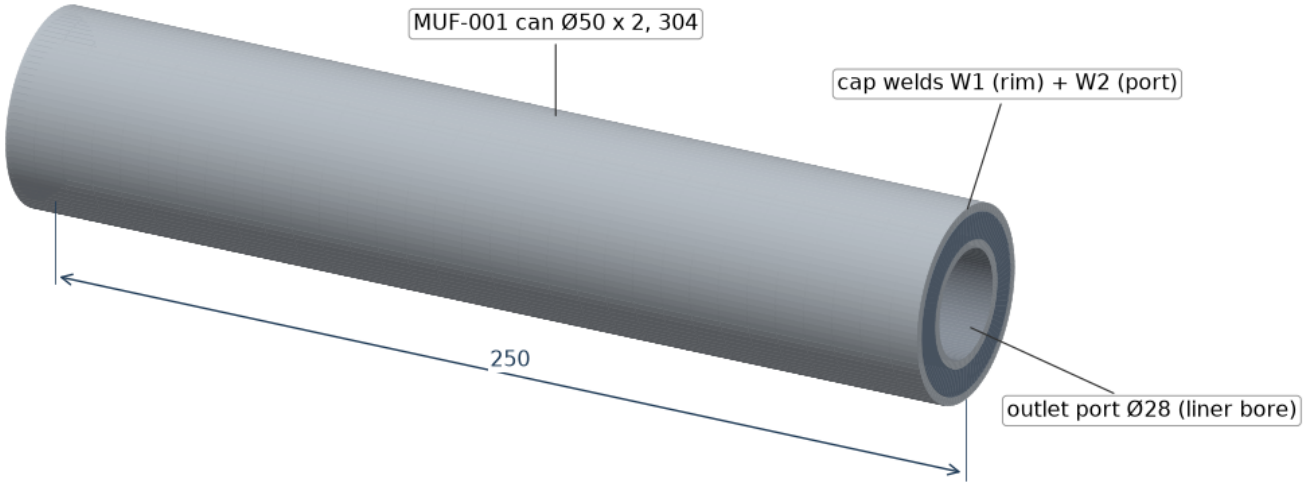
3. 3D Views



Exploded: liner with 112 holes in the middle, baffles below, core fill and can above, caps on the ends. Dashed lines show where each part goes.



Cut-away: quarter removed to show the 4 baffles, the core fill between them and the perforated liner.



Assembled: caps flush with the can ends; the liner bore is the inlet/outlet port.

4. TIG Parameters

Shop machine: 220 V input, 110 A max, AC/DC/pulse, back-purge available.

ID	Joint	DC (DCEN)	Pulse (recommended)	Argon	Back-purge	Filler
W1	End cap OD to can rim	65-85 A	base 40-60 A, peak 100 A, 30-50 Hz	8-10 L/min (gas lens 10-12)	4-6 L/min	ER308L 1/16 in
W2	End cap bore to liner OD (port)	80-100 A	base 60-80 A, peak 100 A, 30-50 Hz	8-10 L/min (gas lens 10-12)	4-6 L/min	ER308L 1/16 in
W3	Baffle ID to liner OD	80-100 A	base 60-80 A, peak 100 A, 30-50 Hz	8-10 L/min (gas lens 10-12)	4-6 L/min	ER308L 1/16 in

- **DCEN for all joints.** AC is for aluminium; don't use it on stainless.
- **Pulse is recommended** on 2 mm 304: lower heat input means less burn-through at hole webs and less sugaring.
- W2 is the same joint as W3 (2 mm liner to 3 mm disc fillet), so it uses the same settings. W1 is an edge/corner
- Peak current never exceeds 100 A, which leaves headroom on the 110 A machine.
- Gas: Ar 8-10 L/min (10-12 with a gas lens); postflow 8-10 s. 2% lanthanated 1/16 in tungsten, #7 cup.
- **Back-purge 4-6 L/min on every hollow joint** (W3 on the liner, W1/W2 at the caps). Foil-tape the ends and
- Tack first (4 per circular joint), then continuous. No preheat for TIG on 304. Interpass < 150 C.
- No W4: the liner is bought tube, so there is no shop seam. With welded tube the factory seam may be drilled through.

Weld map

ID	Joint	Qty	Type / extent	Size
W1	End cap OD to can rim	2	Flush corner (butt-style), full 360, outside	2 (= can wall)
W2	End cap bore to liner OD (port)	2	Fillet, full 360, outside face	2 leg
W3	Baffle ID to liner OD	4	Fillet, full 360, upstream face	2 leg

No W4: the liner is bought tube, so there is no shop seam.

5. BOM + Sourcing

Part	Name	Material	Qty	Notes	Source
MUF-000	Muffler assembly	-	1	Weldment; MUF-000 DXF, weld map, build book	-
MUF-001	Outer can	304 tube 50 OD x 2 wall	1	Cut 250 +/-0.5, square ends, deburr. Measure ID x4.	Online Metals 50x2 304, 60 in length (MTR available) or Amazon 50x2x400 mm (cut to 250)
MUF-002	Perforated liner	304 tube 32 OD x 2 wall (seamless pref.)	1	Cut 250 +/-0.5; 112x DIA 9 radial per drilling spec, using JIG-001; deburr ID+OD	AliExpress 32x2 304 tube, 300 mm (cut to 250) or uxcell 32x2 on Amazon. Fallback rolled 2 mm sheet = 7 holes/row + W4 seam variant (not generated here)
MUF-003	End cap (inlet/outlet port)	304, 3 sheet	2	Laser/plasma+turn: OD 45.8, bore 32.2; port = liner bore	3 mm 304 plate; laser-cut service (SendCutSend, OSH Cut, local shop) from MUF-003/004 DXF
MUF-004	Baffle	304, 3 sheet	4	Same blank as MUF-003	3 mm 304 plate; laser-cut service (SendCutSend, OSH Cut, local shop) from MUF-003/004 DXF
MUF-005	Packing (optional)	Stainless wool + basalt/E-glass muffler packing	1	~30 g; core fill	Motorcycle muffler repack kit
MUF-006	TIG filler rod	ER308L 1/16 in (1.6 mm); ER316L if 316	1	~0.25 kg tube	Welding supply
JIG-001	Drill jig	0.5 mm shim + 38 OD tube + steel block + M6 nut	1	See jig_build.md	Shim stock + scrap

• **Liner** (32 x 2 304 tube): AliExpress 32x2 304 tube, 300 mm (cut to 250) or uxcell 32x2 on Amazon. Fallback rolled 2 mm sheet = 7 holes/row + W4 seam variant (not generated here).

• **Can** (50 x 2 304 tube): Online Metals 50x2 304, 60 in length (MTR available) or Amazon 50x2x400 mm (cut to 250).

• **Discs** (caps + baffles): 3 mm 304 plate; laser-cut service (SendCutSend, OSH Cut, local shop) from MUF-003/004 DXF. Order **after** measuring the tube (step 1).

• US imperial near-equivalents if metric is short: 2 in x 0.083 in (can), 1-1/4 in x 0.083 in = 31.75 x 2.11 (liner);

• 316/316L if the vehicle sees road salt or marine use; then use ER316L filler.

• Filler ER308L 1/16 in (ER316L for 316), 2% lanthanated tungsten, argon.

6. Verification Checklist

Part: 50 x 250 baffle muffler. Builder: _____ Date: _____

Liner (before assembly)

[ ] Hole count = **112** (14 rows x 8). Count by row, tick each row off.

[ ] Hole Ø9 +0.3/-0, round (caliper in two directions).

[ ] Web check: thinnest web between holes >= 1.9 mm (check on the ID; design 2.00). Axial web between rows >= 2 mm (design 3.00).

[ ] No hole within 3 mm of baffle/cap positions (z = 50, 100, 150, 200 and both ends).

[ ] ID and OD deburred, no chips inside.

Fit

[ ] Can ID measured x4 each end: smallest _____ ; disc OD = ID - 0.2 = _____ (design 45.8).

[ ] Cap fit: cap enters can by hand pressure, gap <= 0.3 all round, sits flush with the can end.

[ ] Discs slide on the liner without forcing.

[ ] Baffle spacing: centres at 50, 100, 150, 200 +/-1 mm, square to axis within 0.5 mm at OD.

Welds

[ ] W3 x4, W2 x2, W1 x2 complete, full 360, no cracks/porosity/undercut; colour straw-blue, not grey.

[ ] Soapy-water leak test: plug the outlet, Schrader plug in the inlet, **0.2-0.3 bar (3-4 psi) max**. Brush soap on every W1 and W2 joint: no bubbles in 30 s. (W3 is internal; leaks between chambers don't matter.)

Final

[ ] Rattle check: shake hard by hand; nothing loose.

[ ] Sound check: engine at idle and 3/4 throttle, phone dB meter at 0.5 m, 45 deg from the outlet; record without / with muffler: _____ / _____ dB(A).

[ ] Welds passivated, no rust bloom after 24 h.

Design checks (computed by muffler.py)

Check	Value	Result
Hole count	112 = 14 rows x 8	PASS
Web, within row @ ID (worst)	2.00 mm >= 1.9	PASS
Web, within row @ OD	3.57 mm	PASS
Web, between rows (axial)	3.00 mm >= wall 2	PASS
Web, diagonal (22.5 deg offset) @ ID	4.20 mm >= 1.9	PASS
Open area, whole liner	28.4% on OD (32 mm) basis, target 27-33%; 32.4% on ID (28 mm) flow-path basis	PASS
Hole edge to cap/baffle face	6.25 mm >= 3.0	PASS
Disc OD in can ID	45.80 < 46.00	PASS
Disc bore over liner OD	32.20 > 32.00	PASS
Baffle count 3-5	4	PASS
Max TIG amps (DC / pulse peak) <= welder	100 A <= 110 A	PASS
Guide plate pattern = liner pattern	112 = 112	PASS